Preserving Visual Evidence in Compressed Multimodal Reasoning

Working Title

Preserving Visual Evidence in Compressed Multimodal Reasoning

Problem Statement

SketchThinker-R1 checks key-step preservation in cold-start data, but its training-time style judge sees neither the image nor the question. A concise trace can state only correct steps and still omit a visual premise needed to support its answer.

Proposed Solution

We will test whether supervision that detects missing visual support improves compressed reasoning beyond style or stated-step correctness rewards. Using Qwen2.5-VL-7B-Instruct and 3,000 verifiable geometry and bar-chart problems, we will split by template family before generating traces or image variants and annotate admissible derivations, accepting valid alternative proofs. A sufficient trace must explicitly state correct required visual premises and valid deductions supporting its returned answer; the image checks stated facts without filling omissions. A judge will learn from supported traces, correct-answer traces with premises deleted, false readings, and redundant-step deletions, with independent human checks on held-out problems. Four reward arms—sufficiency, style, stated-step correctness, and zero auxiliary reward—will share compressed cold-start data, initialization, accuracy/format rewards, and a measured policy-training compute ceiling including judge scoring. Common budget conditioning and a length penalty will support comparisons at matched achieved reasoning lengths. We will evaluate answer accuracy, human-rated sufficiency, visual-premise retention, end-to-end latency, and correctness on both members of answer-changing image pairs; nuisance edits will test invariance. Separate ablations will remove judge image access and omission-negative training. Original-trace and same-rewriter length-preserving controls will test rewriting effects. Three training seeds and problem-family confidence intervals will quantify uncertainty, and a judge/throughput pilot will precede policy training.

Key Research Questions

- Under compression, which necessary visual premises disappear, compared with nonvisual steps, after accounting for valid alternative derivations?
- At matched achieved lengths and training compute, does omission-sensitive supervision improve accuracy and explanation sufficiency beyond style and stated-step correctness rewards?
- Do gains persist on answer-changing image pairs and when judge image access or omission-negative training is removed?

Expected Outcome

The study will establish whether explicit visual-support supervision improves the accuracy–length tradeoff and deliver an independently evaluated compression method, including the conditions where it offers no advantage.